

TrES-1b

R-1371

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_130510 · created 2026-08-01T00:47:12+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456422.92011 +/- 0.0007 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1598 +/- 0.0083
Transit depth (Rp/Rs)^2	2.55 +/- 0.27 [%]
Orbital Inclination (inc)	88.6 +/- 0.68
Airmass coefficient 1 (a1)	1.012 +/- 0.01
Airmass coefficient 2 (a2)	-0.007 +/- 0.016
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	#2 - [439, 232]
Optimal Aperture	3.34
Optimal Annulus	15.35
Transit Duration (day)	0.1031 +/- 0.0031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1598 ± 0.0083 vs 0.1358 (deviation 2.67σ)
Duration fit vs published	0.1031 d vs 0.1042 d (deviation 0.32σ)
Mid-time O–C	-2.8 min
Depth SNR	17.8
Residual scatter	1.02 %

Light curve

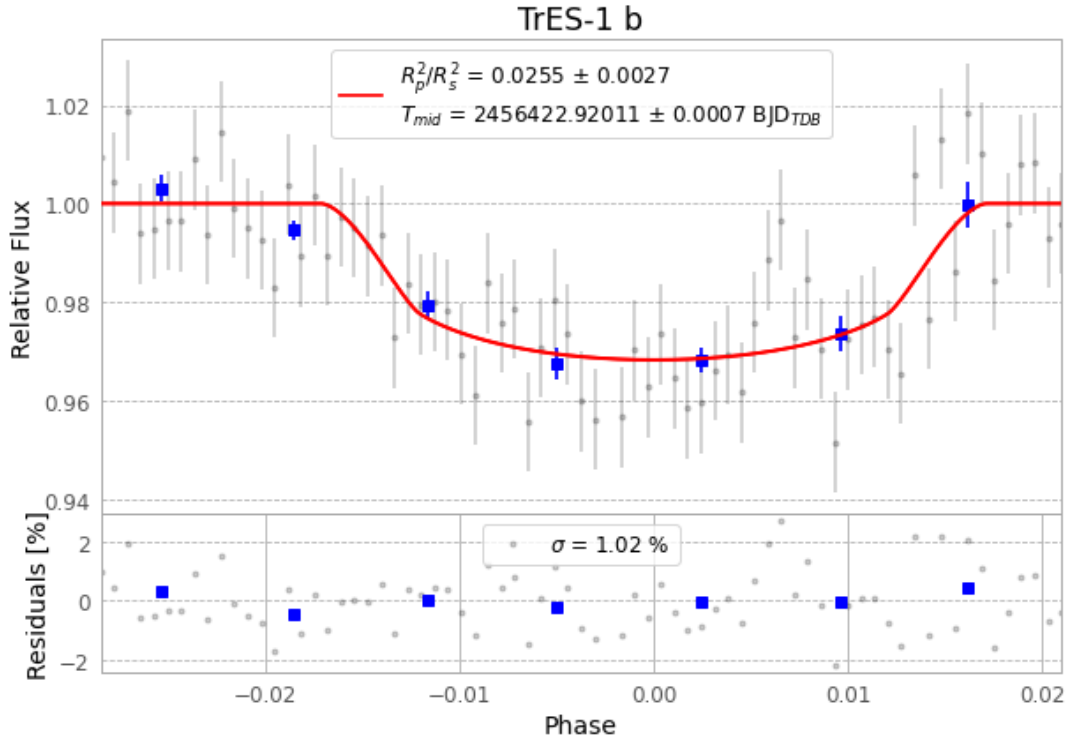


Figure 1 — FinalLightCurve_TrES-1 b_2013-05-10.png (SHA-256 e88787d147c931d3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [369, 226], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 61f67f11ccfd2e1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), TRES-1130510075712.FITS → TRES-1130510113312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-130510.log (e86429aff828...), FinalLightCurve_TrES-1 b_2013-05-10.png (e88787d147c9...), FinalLightCurve_TrES-1 b_2013-05-10.csv (8c4598e112c1...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 00:47Z.